

Practice Sheet: Probability

1. A committee of 4 people is to be appointed from 3 officers of the production department, 4 officers of the purchase department, 2 officers of the sales department and 1 chartered accountant, Find the probability of forming the committee in the following manner:
 - (a) There must be one from each category.
 - (b) It should have at least one from the purchase department.
 - (c) The chartered accountant must be in the committee.
2. A bag contains 7 white, 6 red and 5 black balls. Two balls are drawn at random. Find the probability that they will both be white.
3. Two different digits are chosen at random from the set 1, 2, 3, ..., 8. Show that the probability that the sum of the digits will be equal to 5 is the same as the probability that their sum will exceed 13.
4. If two dice are thrown, what is the probability that the sum is (a) greater than 8, and (b) neither 7 nor 11?
5. A problem in Statistics is given to the three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{3}{4}$, and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved if all of them try independently?
6. A bag contains 17 balls marked with the numbers 1 to 17. A ball is drawn and replaced; a second drawing is then made. What is the probability that
 - (a) the first number drawn is even and the second is odd?
 - (b) the first number is odd and the second is even?How will your result in (a) and (b) be effected if the first ball drawn is not replaced?
7. Six dice are thrown simultaneously. What is the probability that all will show different faces?
8. A bag contains 10 gold and 8 silver coins. Two successive drawings of 4 coins are made such that : (i) coins are replaced before the second trial, (ii) the coins are not replaced before the second trial. Find the probability that the first drawing will give 4 gold and the second 4 silver coins.
9. Three machines X, Y, Z with capacities proportional to 2:3:4 are producing bullets. The probabilities that the machines produce defective are 0.1, 0.2 and 0.1 respectively. A bullet is taken from a day's production and found to be defective. What is the probability that it came from machine X?
10. Two cards are drawn from a deck of 54 playing cards, two of them being jokers. What is the probability that (i) They are both spades? (ii) They are both kings? (iii) At least one of them is a joker? (iv) One of them is a queen and the other either a jack or a joker?
11. In a lottery of 50 tickets numbered 1 to 50, two tickets are drawn simultaneously. Find the probability when: (i) both the tickets drawn are prime numbers, (ii) none of the tickets is prime number.
12. A bag contains 5 white and 7 black balls. Three balls are drawn in succession. What is the probability that all three are white if (i) each ball is replaced before the next one is drawn, (ii) none of these balls is replaced before the next draw?
13. A ball is drawn at random from one of two boxes. The probability of selecting Box A is 0.5, and the probability of selecting Box B is also 0.5. Box A contains 4 red balls and 6 blue balls. Box B contains 7 red balls and 3 blue balls. If the ball drawn is red, find the probability that it was chosen from Box B.
14. A disease affects 1% of a population. A test for the disease is 95% accurate if a person has the disease (true positive) and 90% accurate if the person does not have the disease (true negative). If a person tests positive, what is the probability that they actually have the disease?
15. You have two coins: Coin 1 is fair ($P(\text{Head}) = 0.5$), Coin 2 is biased ($P(\text{Head}) = 0.8$). You pick a coin at random and toss it once. If the result is a Head, what is the probability that you picked the biased coin?